INTELLIGENT COURSE RECOMMENDATION PLATFORM

Data-Driven Learning at Scale

THE PROBLEM

Students worldwide face overwhelming choice paralysis. With millions of courses available across Coursera, Udacity, LinkedIn Learning, and more — each with varying quality, price, and relevance — learners struggle to:

- Identify which courses align with their career goals
- Discover optimal learning paths without wasting time and money
- Find courses that match their skill level and learning pace
- Navigate quality variations and outdated curriculum

Education platforms and course providers, meanwhile, have access to rich course metadata and student engagement data — but lack centralized intelligence to surface recommendations and provide students with data-driven insights.

OUR SOLUTION

We build a production-grade, cloud-native platform with two integrated products:

1. AI-Powered Chatbot

A conversational recommendation engine that understands student goals, skill level, and preferences — then generates personalized course roadmaps backed by real data. Students ask natural questions like

- *"I want to transition to data science from accounting — what should I learn?"*
- *"Which Python courses fit my budget and timeline?"*
- *"Build a learning plan for cloud engineering that gets me job-ready in 6 months."*

The chatbot runs an LLM (via LangChain) that:

- Understands intent from conversational queries
- Generates SQL dynamically to query course and student data
- Constructs personalized learning roadmaps
- Maintains session context for multi-turn conversations

2. Analytics & Insights Platform

A Power BI-powered analytics dashboard for operators, educators, and platform managers to understand course performance at scale:

- Which courses drive the highest completion and job-placement rates?
- What skill combinations are most valuable?
- How do student cohorts and learning paths correlate with outcomes?
- What gaps exist in course availability?

IMPACT

- **For Students:** Personalized learning paths eliminating wasted time and money
- **For Platforms:** Data-driven insights to optimize course curation and pricing
- **For EdTech Industry:** Reusable architecture deployable across any education provider
- **For Investors:** Defensible product with dual monetization paths (B2C & B2B)

PLATFORM ARCHITECTURE

01 INGEST	Kaggle · Course provider APIs · CAPMAS data · Market pricing feeds
02 STORE	Azure Data Lake Gen2 · Raw zone (landing data) · Curated zone (cleaned data)
03 PROCESS	dbt + DuckDB · Schema standardization · Deduplication · Enrichment & joins
04 WAREHOUSE	DuckDB Analytical Database · Star schema · Fast OLAP queries
05 AI LAYER	LangChain · LLM (Claude / GPT-4) · SQL generation · Recommendation logic
06 SERVE	Web UI (Chatbot) · Power BI dashboards · MongoDB session store (conversation context)

Data Flow Summary

- Course metadata ingested from Kaggle/APIs → Raw Zone (Data Lake)
- Cleaned & modelled via dbt → Analytical Database (DuckDB)
- Student query → LLM generates SQL → Executes on DuckDB → Returns results
- Modelled data from DuckDB → Power BI → Dashboards for analysts

TECHNOLOGY STACK

Data Engineering	dbt, DuckDB, Azure Data Lake, Apache Airflow, Python
AI & LLM	LangChain, Claude API, MongoDB (session store)
Analytics	Power BI, SQL
DevOps & Frontend	Docker, GitHub Actions, React/Node.js

TEAM STRUCTURE

Role	Responsibilities
Data Ingestion	API integrations, data validation, raw data pipelines
Orchestration	Apache Airflow DAGs, pipeline scheduling, monitoring
Warehouse Design	Star schema, dimensional modelling, dbt development
Data Processing	dbt transformations, data quality checks, enrichment
AI & LLM	LangChain integration, prompt engineering, SQL generation
BI & DevOps	Power BI dashboards, Docker, CI/CD, infrastructure

KEY FEATURES

For Students (Chatbot)

- Natural language interface — no technical knowledge required
- Personalized course recommendations based on goals, budget, timeline
- Auto-generated learning roadmaps with sequencing logic
- Multi-turn conversation with session persistence

For Operators (Analytics)

- Real-time Power BI dashboards showing course performance
- Skill-to-outcome mapping and career progression tracking
- Cohort analysis and learning path effectiveness
- Data-driven curriculum gaps and recommendations

Ready to transform education through data-driven personalization.

Let's build the future of learning together.